# Investigating accelerometer accuracy by integrating twice

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# Contents

**1 Introduction** **2**

**2 How to find displacement from acceleration graph** **3**

**3 Modeling** **7**

**4 Integrating to velocity** **9**
4.1 Integrating to displacement . . . . . 12

**5 Comparison** **13**

**6 Video** **14**

**7 Appendix** **14**
7.1 Python code . . . . . 14

# 1 Introduction

In Vex robotics, there are sensors for quickly measuring locations for accurate repeated robotic automationos. And the most useful one is an IMU. An inertial measurement unit (IMU) is a sensor that measures acceleration at a high rate. Whenever you need to know the precise location of something quickly, double integrating accelerometer data outpaces GPS solutions. They rely on the property that the area under a accelerometer graph can provide enough data used to compute the displacement of the object. In this investigation we will explore numerical and analytical integration of the accuracy of accelerometer data recorded on a phone accelerometer and comparing with video recorded displacement.

## 2 How to find displacement from acceleration graph

Recall that in kinematics acceleration is defined as the derivative of velocity

$$a = \frac{\mathrm{d}v}{\mathrm{d}t} \tag{1}$$

To find the derivative from first principals, we use limits:

$$\lim_{h \to 0} \frac{v(t+h) - v(t)}{h} \tag{2}$$

and $v(t+h) - v(t)$ essentially means the change in velocity over a short period of time $h$. And as $h$ approaches 0, it becomes an increasingly more accurate representation of the derivative of acceleration, velocity. And this means that for our data to be accurate, the sample rate has to be high.

Recall the fact that velocity is the integral of acceleration and displacement is the integral of velocity. Integral is the area under a curve. For continuous functions, they are an approximateted by dividing the function into many "rectangles" like shown below where the height of a specific rectangle at a specific point of the function is the y value of the function like shown below.

Figure 1: Function area approximated with rectangles

As the width of the rectangles becomes smaller and smaller, the accuracy increases. So in terms of collecting our data, having data with higher sample rate then it would become more accurate.

But instead of using rectangles to find the area under the curve, we can just use basic geometric equations like the equation for finding the area of a triangle and rectangle.

$$A_{triangle} = \frac{1}{2} b_t \cdot h_t \tag{3}$$

Figure 2: The formula finding the area of a triangle

$$A_{rectangle} = b_r \cdot h_r \tag{4}$$

Figure 3: Formula for the area of a rectangle

the subscripts r and t indicate which base we are refering to (for instance it is the triangles rather than the rectangle's)

Lets say we have two data points, the second data point's y-value could either be higher, lower, or equal to the y-value of the first point (shown below). And the numerical integral is essentially the area under between these two points, which can be calculated with the previous geometric equations illustrated below labeled with the equations

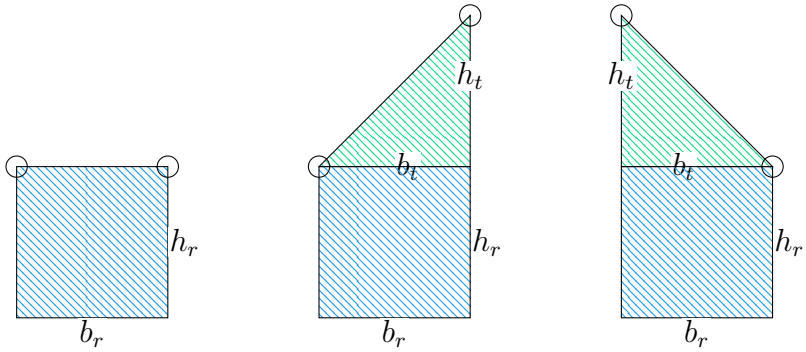



Figure 4: The three conditions of integration

But of course our data isn't just two points, but for the first two points, we find the area, and then the integral function's will use that area as the y-value for the new point. And the point's x-value should be in between the two points that we calculated the area with. And if we have another point, the integral function's corresponding new point's y-value will be the previous point's y-value plus the area between the new point and the previous point, and once again the x-value will be in between the two points in which the area is calculated from. So this is the data we collected from swinging the phone right to left 5 times. Positive means to the e Shown in Figure 5

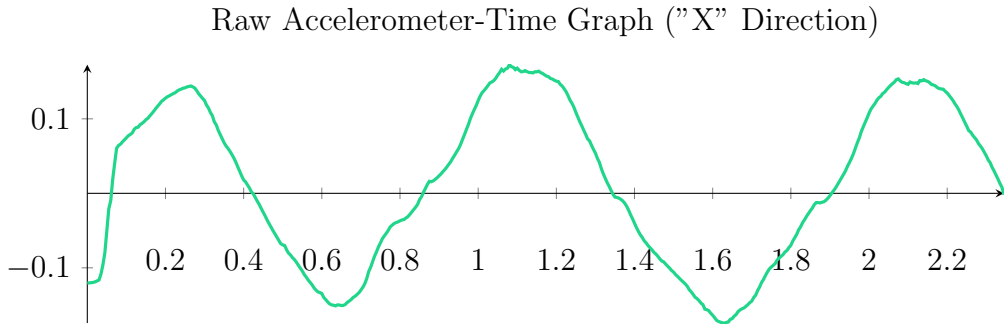



Figure 5: Raw accelerometer data on the X axis

So now have the tools to numerically integrate a piece of data, giving us a velocity-time graph. Shown below in Figure 6

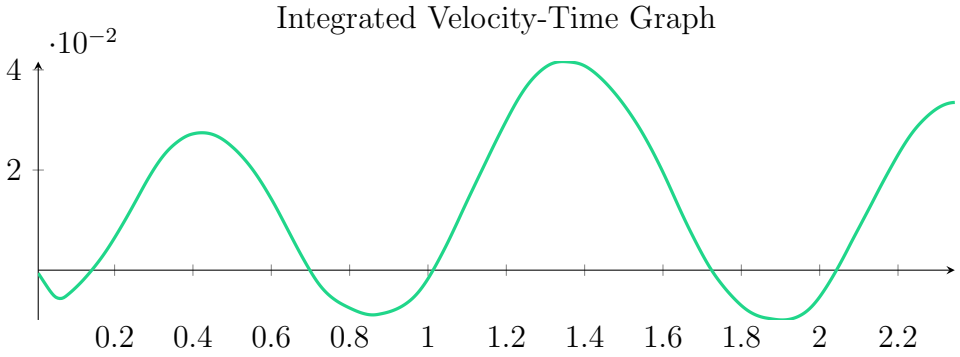



Figure 6: Integrated accelerometer data (velocity)

In theory, if we integrate again, we can finally get a displacement-time graph. But the velocity graph feels a little off because the area is mostly on the upper side of the x-axis, meaning if we were to integrate it, the displacement would start going towards the right rather than what actually happened: going right and left 5 times and then stopping.

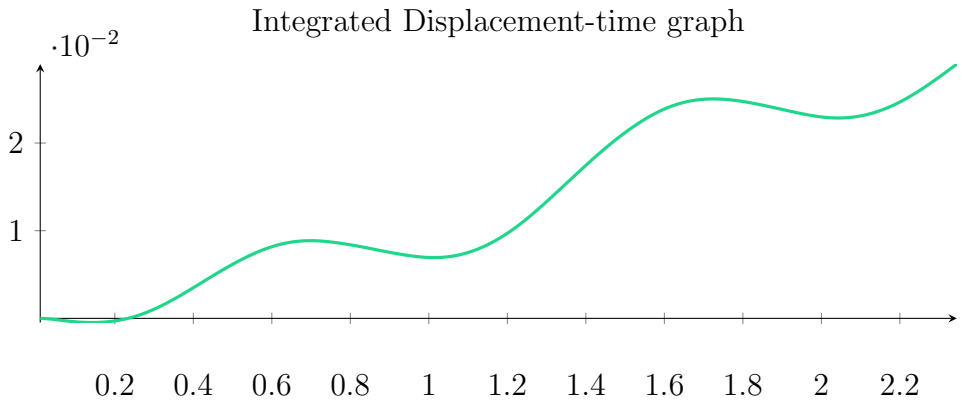



Figure 7: Integrated accelerometer data (Displacement)

One way I thought of to correct this is to center the accelerometer-time graph before integrating it into velocity. As you can see below

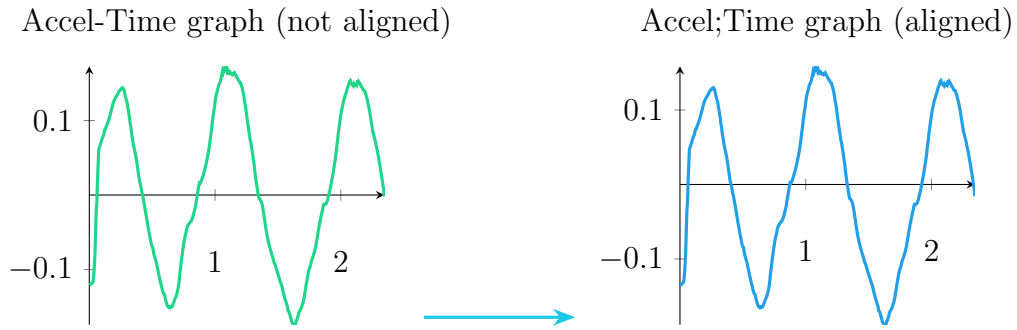



Figure 8: Raw accelerometer data on the X axis

And now we try integrating again:

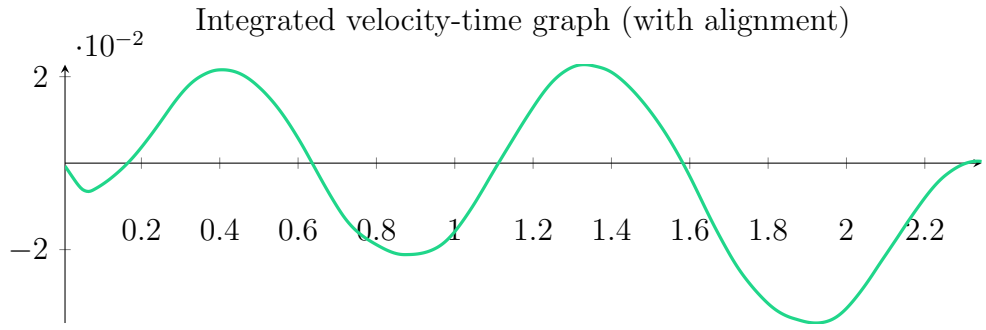



Figure 9: Integrated velocity (aligned)

Ok, much better. We should probably center it again before integrating it once more to get the displacement-time graph.

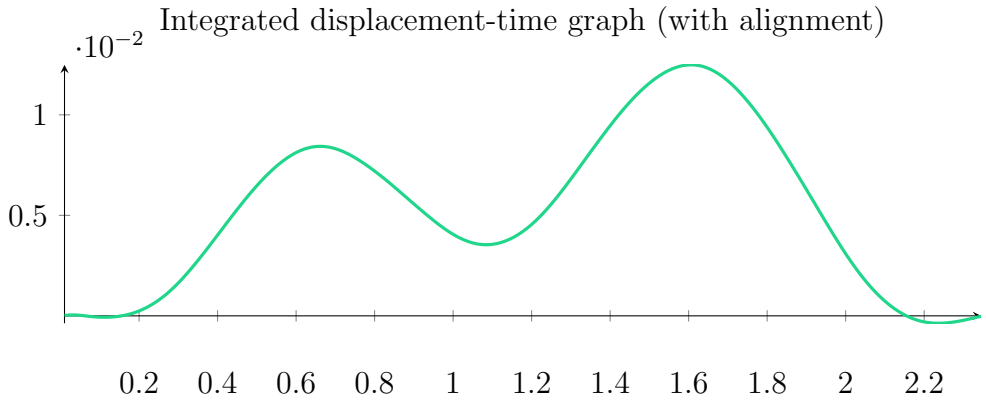



Figure 10: Integrated displacement (aligned)

# 3 Modeling

I decided to compare integrating numerically against analytically; I want to know if there is a difference if I first modeled it with a sine equation and then integrated the equation algebraicly rather than numerically using geometry like I did before.

Desmos has a tool for fitting data into many types of equations. In particular, we are looking at a sinulsoidal regression

$$y = a\sin(bx + c) + d \tag{5}$$

Plugging into desmos for sinusoidal regression we get:

$$y = 0.14272 \cdot \sin\left(6.6486x + 0.216978\right) - 0.00389886 \tag{6}$$

seen in figure 11.

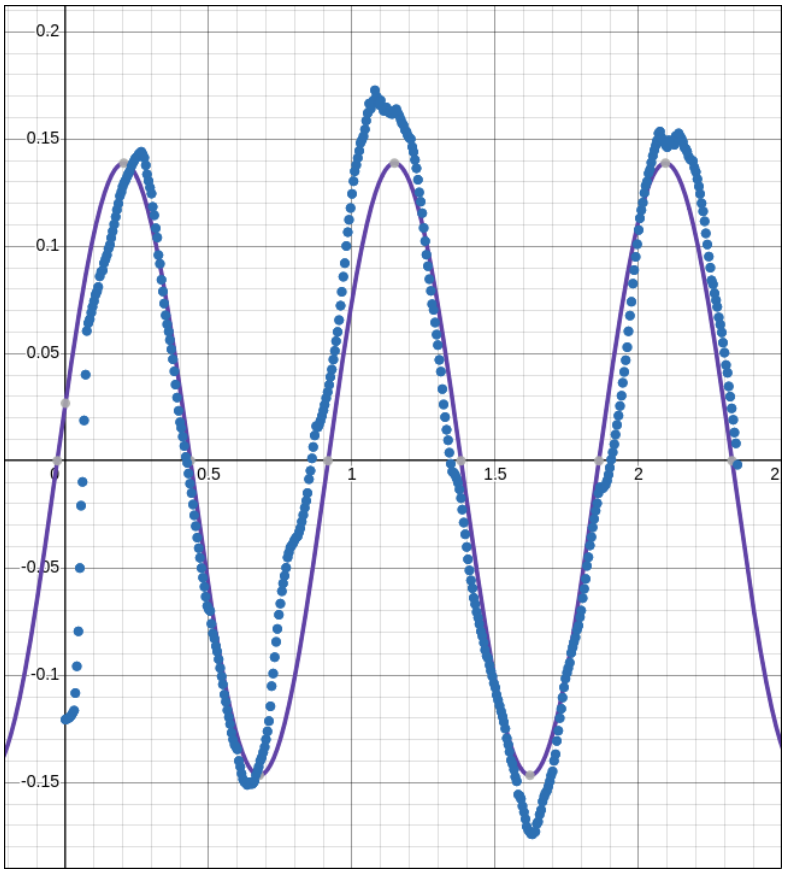



Figure 11: graph of our accelerometer data and the fitted curve

Intuitively, the last d term represents sensor drift, if we integrated $sin(x)+1$ to find velocity $v(t)$:

$$v(t) = \int sin(x) + 1 \, \mathrm{d}x \tag{7}$$

$$= -cos(x) + x + c \tag{8}$$

then, it's graph

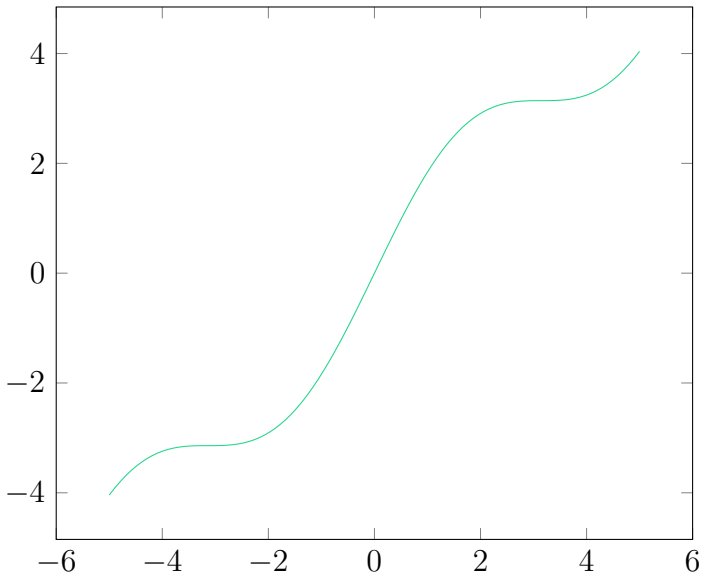



Figure 12: graph of $\sin(x) + x$

would just go up indefinitely, which wouldn't make sense especially when I moved my the phone back and forth and ended at rest because if the velocity never comes back down to zero, it would never come back to rest. If we just remove the d term in $a\sin(bx + c) + d$ then, its integration won't have any drift. Removing the d term in this sinusoidal regression before integration is akin to centering the accelerometer data before integration in the numerical method.

# 4 Integrating to velocity

We will need to integrate once to get velocity:

$$v(t) = \int (a \cdot \sin(bt + c) - d)\,\mathrm{d}t \tag{9}$$

where a, b, c, and d are constants. Rearranged, we get

$$v(t) = a\int \sin(bt + c)\,\mathrm{d}t - \int d\,\mathrm{d}t \tag{10}$$

Using u-substitution

$$u = bt + c \tag{11}$$

$$\frac{\mathrm{d}u}{\mathrm{d}t} = b \tag{12}$$

$$\mathrm{d}t = \frac{\mathrm{d}u}{b} \tag{13}$$

plugging in u and dt:

$$v(t) = a\int \sin(bt + c)\,\mathrm{d}t - \int d\,\mathrm{d}t \tag{14}$$

$$= a\int \sin(u)\frac{1}{b}\,\mathrm{d}u - \int d\,\mathrm{d}t \tag{15}$$

$$= \frac{a}{b}\int \sin(u)\,\mathrm{d}u - \int d\,\mathrm{d}t \tag{16}$$

$$= -\frac{a}{b}\cos(u) - \int d\,\mathrm{d}t \tag{17}$$

$$= -\frac{a}{b}\cos(bt + c) - dt + C_1 \tag{18}$$

we get our velocity time equation

$$v(t) = -\frac{a}{b}\cos(bt + c) - dt + C_1 \tag{19}$$

In kinematics, the constant $C_1$ is the initial velocity, and since we know that our initial velocity is 0, we can ignore $C_1$ and set it to zero...is what I would say if I didn't pay enough attention. Because if we find the constant $C_1$ of our modeled function when initial velocity $v(0)$ is actually $= 0$

$$v(0) = -\frac{a}{b}\cos((0)\,b + c) + C_1 = 0 \tag{20}$$

$$\frac{a}{b}\cos(c) = C_1 = 0.0209628448456 \tag{21}$$

$$\tag{22}$$

It is not actually zero! That's because $C_1$ is dependent on the other terms a, b, and c. But still, it doesn't mean that $C_1$ doesn't depend on initial velocity, it may not depend on d, but d isn't the only value that determines initial velocity.

As we mentioned before, removing d from our sinusoidal regression removes a certain amount of drift; particularly to the extent that our velocity value can still return to zero (come to rest) no matter how large the value t (however long time has passed). But still, even without d: in order for $a(t) = a\sin(bt + c)$'s integrated velocity function $-\frac{a}{b}\cos(bt + c) + C_1$ to end at zero, we must also define the range of the function such that we end at $v(t) = -\frac{a}{b}\cos(bt + c) + C_1 = 0$. In this case, we shouldn't focus on $a$ because that is solely the amplitude while b and c determine the horizontal characteristics of the sine wave function such as horizontal compression and phase shift. and phase shift in particular is responsible for where on the wave we start, and even if the vertical shift is 0, we could still start at the crest of the sine function, which would mean our initial velocity isn't zero.

So now we figured out that a $d > 0$ will result in the velocity function $v(t) = -\frac{a}{b}\cos(bt + c) - dt + C_1$ to never reach zero as time $t$ increases, and setting $d = 0$ will make it so we can still end up at zero. It doesn't exactly still imply that our displacement will go back to zero.

. Regardless, we know if we continue integrating with an additional constant, its bound to add drift, because $\int c_1 dx = c_1 x + c$ and as x increases, the y value also goes up. But this time we have no choice but to keep it because,

and graphing it with the correct values we get the following graph

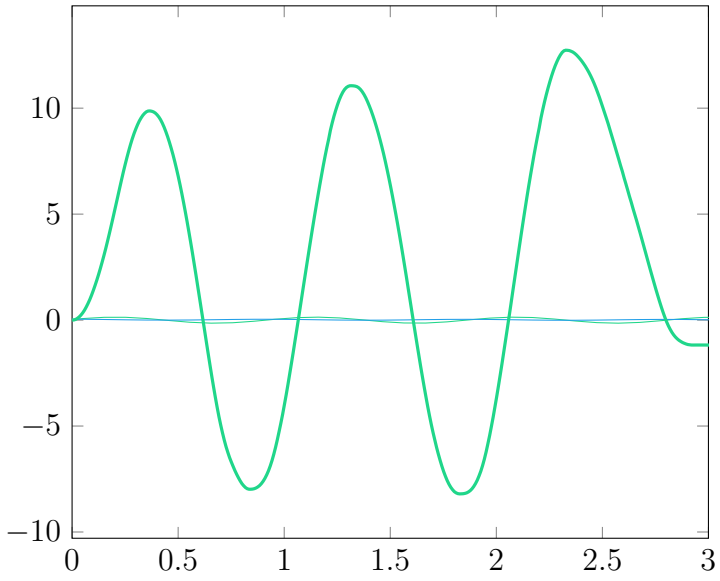



Figure 13: Velocity-time continuous function

Where green is the original function and blue is the integrated function and the orange is the actual recorded displacement.

For now, we will keep in mind that $C_1 = 0.0209628448456$ but continue using $C_1$ throughout integration for convenience. To find the displacement function with respect to time we integrate again.

## 4.1 Integrating to displacement

$$s(t) = \int \left(-\frac{a}{b}\cos(bt+c) - dt + C_1\right) \mathrm{d}t \tag{23}$$

$$s(t) = -\frac{a}{b}\int \cos(bt+c)\,\mathrm{d}t - \int dt\,\mathrm{d}t + \int C_1\,\mathrm{d}t \tag{24}$$

$$s(t) = -\frac{a}{b}\int \cos(bt+c)\,\mathrm{d}t - \int dt\,\mathrm{d}t + \int C_1\,\mathrm{d}t \tag{25}$$

$$\tag{26}$$

Using the same u-substitution:

$$s(t) = -\frac{a}{b}\int \cos(u)\frac{\mathrm{d}u}{b} - \int dt\,\mathrm{d}t + \int C_1\,\mathrm{d}t \tag{27}$$

$$s(t) = -\frac{a}{b^2}\sin(u) - \int dt\,\mathrm{d}t + \int C_1\,\mathrm{d}t + C_2 \tag{28}$$

$$s(t) = -\frac{a}{b^2}\sin(bt+c) - \int dt\,\mathrm{d}t + \int C_1\,\mathrm{d}t + C_2 \tag{29}$$

$$s(t) = -\frac{a}{b^2}\sin(bt+c) - \frac{dt^2}{2} + C_1 t + C_2 \tag{30}$$

$$\tag{31}$$

and since this data is filmed as relative displacement from original point (seen as 0) we need to find the value of $C_2$ such that $s(0) = 0$. [Ai said wording was bad, I don't understand why]

$$s(0) = -\frac{a}{b^2}\sin(b\,(0)+c) - \frac{d\,(0)^2}{2} + C_1\,(0) + C_2 = 0 \tag{32}$$

$$0 = -\frac{a}{b^2}\sin(b\,(0)+c) + C_2 \tag{33}$$

$$\frac{a}{b^2}\sin(b\,(0)+c) = C_2 = 0.000695067592524 \tag{34}$$

$$\tag{35}$$

Again we will keep $C_2 = 0.000695067592524$ in mind. Resulting in a final function $s(t)$:

$$s(t) = -\frac{a}{b^2}\sin(bt + c) - \frac{dt^2}{2} + C_1 t + C_2 \tag{36}$$

drawing a graph of progressive integration is seen in figure 14

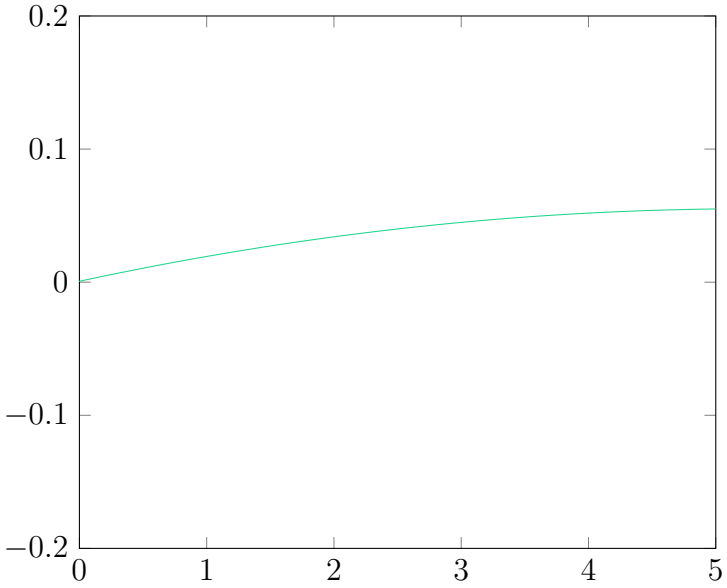



Figure 14: graph of the model being integrated twice

# 5 Comparison

We used two methods that both tries to find the displacement-time graph using a given accelerometer-time data, numerically and analytically. For the numerical approach, we have a lot of points that make up a graph when you connect the points. And since there are points, we can use geometry (although it is a lot of geometry) to find the area under a curve. We centered the data by subtracting each point by the mean of every point's y-value before integrating it numerically to reduce drift during integrated graphs. For the Analytical approach, we fitted a sinusoidal regression on the same accelerometer-time data and then we used algebra and integral calculus to integrate twice to find the displacement-time graph. Here is the both of them

in one graph. The orange is the numerical method while the blue one is the analytical method.

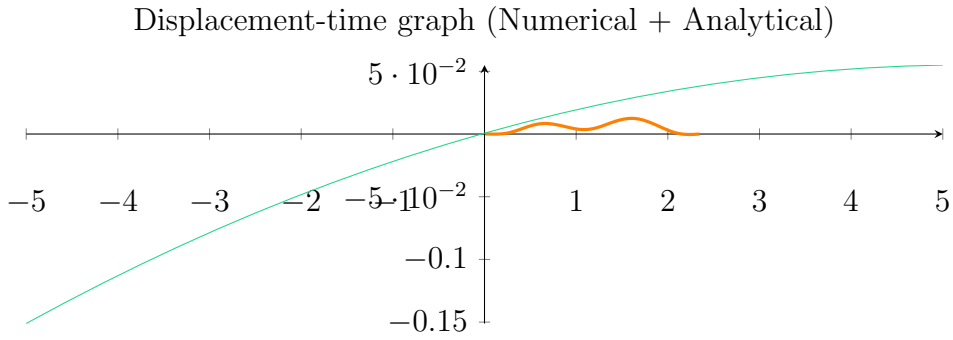



Figure 15: Comparison between integration methods

# 6 Video

We have a displacement/time graph from video. However it is a different sampling rate (120hz) than the recorded datapoints (100hz). Hence we will attempt to align by drawing a line between the points in the recorded accelerometer data, and interpolate the value.

# 7 Appendix

## 7.1 Python code

```
dt = final_time - initial_time
dx_accel = final_accel - initial_accel
dx_vel = (dt * initial_accel) + ((1/2) * dt * dx_accel)
# triangle area + square area = area under accel graph = change in velocity
```

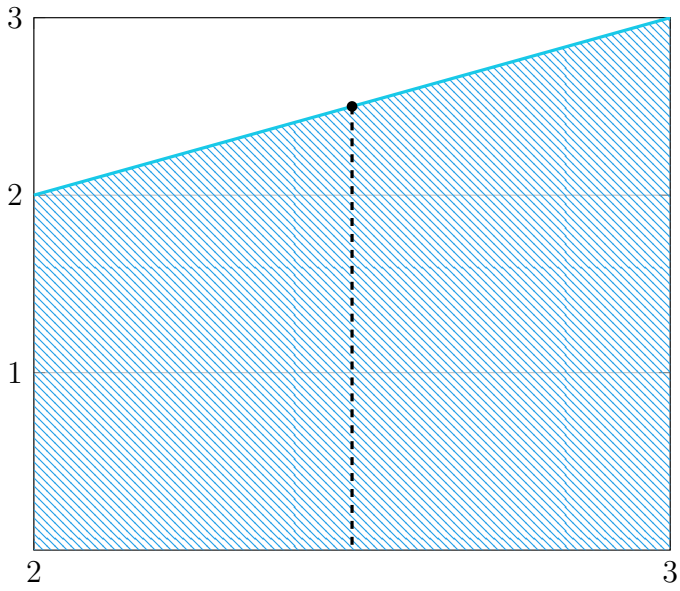



Figure 16: Illustration of interpolating values from two datapoints $x = 2, 3$